

Exercise 13.1

1. Arshad rolls a dice, with sides labelled L, M, N, O, P, U . What is the probability that the dice lands on a consonant?

When a dice is rolled, the sample space will be:

$$S = \{L, M, N, O, P, U\}$$

$$n(S) = 6$$

Let A be an event getting consonant

$$A = \{L, M, N, P\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{6}$$

$$P(A) = \frac{2}{3}$$

2. Shazia throws a pair of fair dice. What will be the probability of getting:

(i) Sum of dots is at least 4.

(ii) Product of both dots is between 5 to 10.

(iii) The difference between both the dots is equal to 4.

(iv) Number at least 5 on the first dice and number at least 4 on the second dice.

When two dice are thrown, the sample space will be:

$$S = \{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,1), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$n(S) = 36$$

(i) Let A be the event getting the sum is at least 4.

$$A = \{(1,3), (1,4), (1,5), (1,6), (2,2), (2,3), (2,4), (2,5), (2,6), (3,1), (3,2), (3,3), (3,4), (3,5), (3,6), (4,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1), (5,2), (5,3), (5,4), (5,5), (5,6), (6,1), (6,2), (6,3), (6,4), (6,5), (6,6)\}$$

$$n(A) = 33$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{33}{36}$$

$$P(A) = \frac{11}{12}$$

(ii) Let B be the event getting the product is between 5 and 10.

$$B = \{(1,5), (1,6), (2,3), (2,4), (2,5), (3,2), (3,3), (4,2), (5,1), (5,2), (6,1)\}$$

$$n(B) = 11$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{11}{36}$$

(iii) Let C be the event getting the difference between both the dots is equal to 4.

$$C = \{(1,5), (2,6), (5,1), (6,2)\}$$

$$n(C) = 4$$

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{4}{36}$$

$$P(C) = \frac{1}{9}$$

(iv) Let D be the event getting number at least 5 on the first dice and number at least 4 on the second dice.

$$D = \{(5,4), (5,5), (5,6), (6,4), (6,5), (6,6)\}$$

$$n(D) = 6$$

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{6}{36}$$

$$P(D) = \frac{1}{6}$$

3. One alphabet is selected at random from the word "MATHEMATICS". Find the probability of getting: (i) vowel (ii) consonant (iii) an E (iv) an A (v) not M (vi) not T

Sample space is

$$S = \{M, A, T, H, E, M, A, T, I, C, S\}$$

$$n(S) = 11$$

(i) Let A be the event getting vowel.

$$A = \{A, E, A, I\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{4}{11}$$

(ii) Let B be the event getting consonant.

$$B = \{M, T, H, M, T, C, S\}$$

$$n(B) = 7$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{7}{11}$$

(iii) Let C be an event getting of getting an E .

$$C = \{E\}$$

$$n(C) = 1$$

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{1}{11}$$

(iv) Let D be the event getting an A .

$$D = \{A, A\}$$

$$n(D) = 2$$

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{2}{11}$$

(v) Let E be the event getting M .

$$E = \{M, M\}$$

$$n(E) = 2$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{2}{11}$$

To find probability not getting M , we have

$$P(E') = 1 - P(E)$$

$$P(E') = 1 - \frac{2}{11}$$

$$P(E') = \frac{11 - 2}{11}$$

$$P(E') = \frac{9}{11}$$

(vi) Let F be the event getting T .

$$F = \{T, T\}$$

$$n(F) = 2$$

$$P(F) = \frac{n(F)}{n(S)}$$

$$P(F) = \frac{2}{11}$$

To find probability not getting T , we have

$$P(F') = 1 - P(F)$$

$$P(F') = 1 - \frac{2}{11}$$

$$P(F') = \frac{11 - 2}{11}$$

$$P(F') = \frac{9}{11}$$

4. Aslam rolled a dice. What is the probability of getting the numbers 3 or 4? Also find the probability of not getting the numbers 3 or 4.

When a dice is rolled, the sample space will be:

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$n(S) = 6$$

Let A be an event getting the numbers 3 and 4

$$A = \{3, 4\}$$

$$n(A) = 2$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{2}{6}$$

$$P(A) = \frac{1}{3}$$

To find probability not getting the numbers 3 and 4, we have

$$P(A') = 1 - P(A)$$

$$P(A') = 1 - \frac{1}{3}$$

$$P(A') = \frac{3 - 1}{3}$$

$$P(A') = \frac{2}{3}$$

5. Abdul Hadi labelled cards from 1 to 30 and put them in a box. He selects a card at random. What is the probability that selected card contains:

- (i) the number 25
- (ii) number between 17 to 22
- (iii) number at least 20
- (iv) number not 27 and 29
- (v) number not between 12 to 15

Sample space is

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$$

$$n(S) = 30$$

(i) Let A be the event getting the number 25.

$$A = \{25\}$$

$$n(A) = 1$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{1}{30}$$

(ii) Let B be the event getting number between 17 to 22.

$$B = \{17, 18, 19, 20, 21, 22\}$$

$$n(B) = 6$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{6}{30}$$

$$P(B) = \frac{1}{5}$$

(iii) Let C be an event of getting number at least 20.

$$C = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30\}$$

$$n(C) = 11$$

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{11}{30}$$

(iv) Let D be the event getting numbers 27 and 29.

$$D = \{27, 29\}$$

$$n(D) = 2$$

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{2}{30}$$

$$P(D) = \frac{1}{15}$$

To find probability not getting numbers 27 and 29, we have

$$P(D') = 1 - P(D)$$

$$P(D') = 1 - \frac{1}{15}$$

$$P(D') = \frac{15 - 1}{15}$$

$$P(D') = \frac{14}{15}$$

(v) Let E be the event getting number between 12 to 15.

$$E = \{12, 13, 14, 15\}$$

$$n(E) = 4$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{4}{30}$$

$$P(E) = \frac{2}{15}$$

To find probability not getting number between 12 to 15, we have

$$P(E') = 1 - P(E)$$

$$P(E') = 1 - \frac{2}{15}$$

$$P(E') = \frac{15 - 2}{15}$$

$$P(E') = \frac{13}{15}$$

6. The probability that Ayesha will pass the examination is 0.85. What will be the probability that Ayesha will not pass the examination?

Let A be the event that Ayesha will pass the examination

$$P(A) = 0.85$$

To find probability that Ayesha will not pass the examination, we have

$$P(A') = 1 - P(A)$$

$$P(A') = 1 - 0.85$$

$$P(A') = 0.15$$

7. Taabish tossed a fair coin and rolled a fair dice once. Find the probability of the following events:

- (i) tail on coin and at least 4 on dice
- (ii) head on coin and the number 2,3 on dice
- (iii) head and tail on coin and the number 6 on dice
- (iv) not tail on coin and the number 5 on dice
- (v) not head on coin and the number 5 and 2 on dice

If tossed a fair coin and rolled a fair dice once then sample space is

$$S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$$

$$n(S) = 12$$

(i) Let A be the event getting the tail on coin and at least 4 on dice.

$$A = \{(T, 4), (T, 5), (T, 6)\}$$

$$n(A) = 3$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{3}{12}$$

$$P(A) = \frac{1}{4}$$

(ii) Let B be the event getting head on coin and the number 2,3 on dice.

$$B = \{(H, 2), (H, 3)\}$$

$$n(B) = 2$$

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{2}{12}$$

$$P(B) = \frac{1}{6}$$

(iii) Let C be an event getting head and tail on coin and the number 6 on dice

$$C = \{(H, 6), (T, 6)\}$$

$$n(C) = 2$$

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{2}{12}$$

$$P(C) = \frac{1}{6}$$

(iv) Let D be the event getting tail on coin and the number 5 on dice.

$$D = \{(H, 5)\}$$

$$n(D) = 1$$

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{1}{12}$$

To find probability getting not tail on coin and the number 5 on dice, we have

$$P(D') = 1 - P(D)$$

$$P(D') = 1 - \frac{1}{12}$$

$$P(D') = \frac{12 - 1}{12}$$

$$P(D') = \frac{11}{12}$$

(v) Let E be the event getting head on coin and the number 5 and 2 on dice.

$$E = \{(H, 5), (H, 2)\}$$

$$n(E) = 2$$

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{2}{12}$$

$$P(E) = \frac{1}{6}$$

To find probability getting not head on coin and the number 5 and 2 on dice, we have

$$P(E') = 1 - P(E)$$

$$P(E') = 1 - \frac{1}{6}$$

$$P(E') = \frac{6 - 1}{6}$$

$$P(E') = \frac{5}{6}$$

8. A card is selected at random from a well shuffled pack of 52 playing cards. What will be the probability of selecting:

(i) a queen

(ii) neither a queen nor a jack

$$\text{Total number of cards} = 52$$

$$n(S) = 52$$

(i) Let Q be the event of getting a queen, so

$$n(Q) = 4$$

$$P(Q) = \frac{n(Q)}{n(S)}$$

$$P(Q) = \frac{4}{52}$$

$$P(Q) = \frac{1}{13}$$

(ii) Let R be the event of getting a queen or jack, so

$$n(R) = 8$$

$$P(R) = \frac{n(R)}{n(S)}$$

$$P(R) = \frac{8}{52}$$

$$P(R) = \frac{2}{13}$$

To find probability of having neither a queen nor a jack, we have

$$P(R') = 1 - P(R)$$

$$P(R') = 1 - \frac{2}{13}$$

$$P(R') = \frac{13 - 2}{13}$$

$$P(R') = \frac{11}{13}$$

9. A card is chosen at random from a pack of 52 playing cards. Find the probability of getting: (i) a jack (ii) no diamond

$$\text{Total number of cards} = 52$$

$$n(S) = 52$$

(i) Let J be the event of getting a jack, so

$$n(J) = 4$$

$$P(J) = \frac{n(J)}{n(S)}$$

$$P(J) = \frac{4}{52}$$

$$P(J) = \frac{1}{13}$$

(ii) Let D be the event of getting a diamond, so

$$n(D) = 13$$

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$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{13}{52}$$

$$P(D) = \frac{1}{4}$$

To find the probability of not getting diamond, we have

$$P(D') = 1 - P(D)$$

$$P(D') = 1 - \frac{1}{4}$$

$$P(D') = \frac{4-1}{4}$$

$$P(D') = \frac{3}{4}$$

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